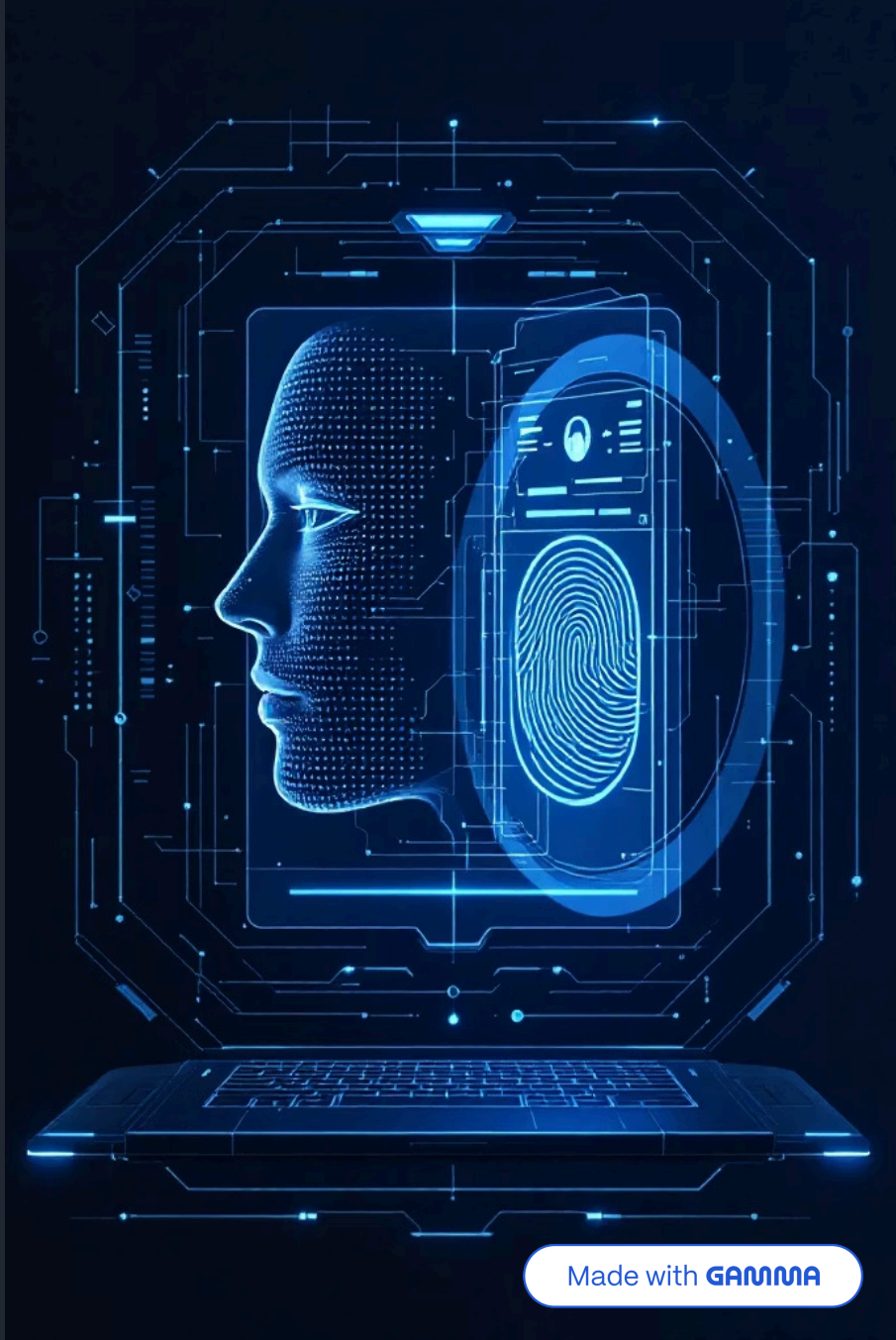
Secure ID Check: Using Face and Fingerprint

Presented by: Himanshu Shekhar

College: Lovely Professional University

Batch : 2023 to 2027

Guide: [Guide Name]



Made with GAMMA

Introduction to Biometric Authentication

What is Biometric Authentication?

Biometrics prove who you are by checking unique things about your body or how you behave. It's about "who you are," not "what you know."

Why Traditional Passwords Fail

Passwords and PINs can be easily guessed or stolen. This makes them risky for people and computer systems.

Everyday Biometrics

Biometrics are already a part of daily life. You use them to unlock your phone, at airport security, and for national IDs like Aadhaar.

Made with GAMMA

Ways We Use Biometrics

OUR FOCUS

Face Scan

Finds people by looking at unique face shapes. Our project uses this.

Fingerprint Scan

Checks who you are by using special lines on your fingertips. This is also a main part of our system.

Eye Scan (Iris)

Uses the unique patterns in your eye's iris for very exact identification. We might add this in the future.

Voice Scan

Confirms who you are by listening to your voice and how you speak. This is another area for future growth.

Our project mainly uses Face and Fingerprint Scans. We know that Eye and Voice Scans could be good to add later.

Made with GAMMA

The Problem: Weaknesses in Today's Security

Weak Passwords

The passwords we use are often weak, used again, or easy to guess. This makes them an easy target for attackers.

Stolen Codes (OTPs)

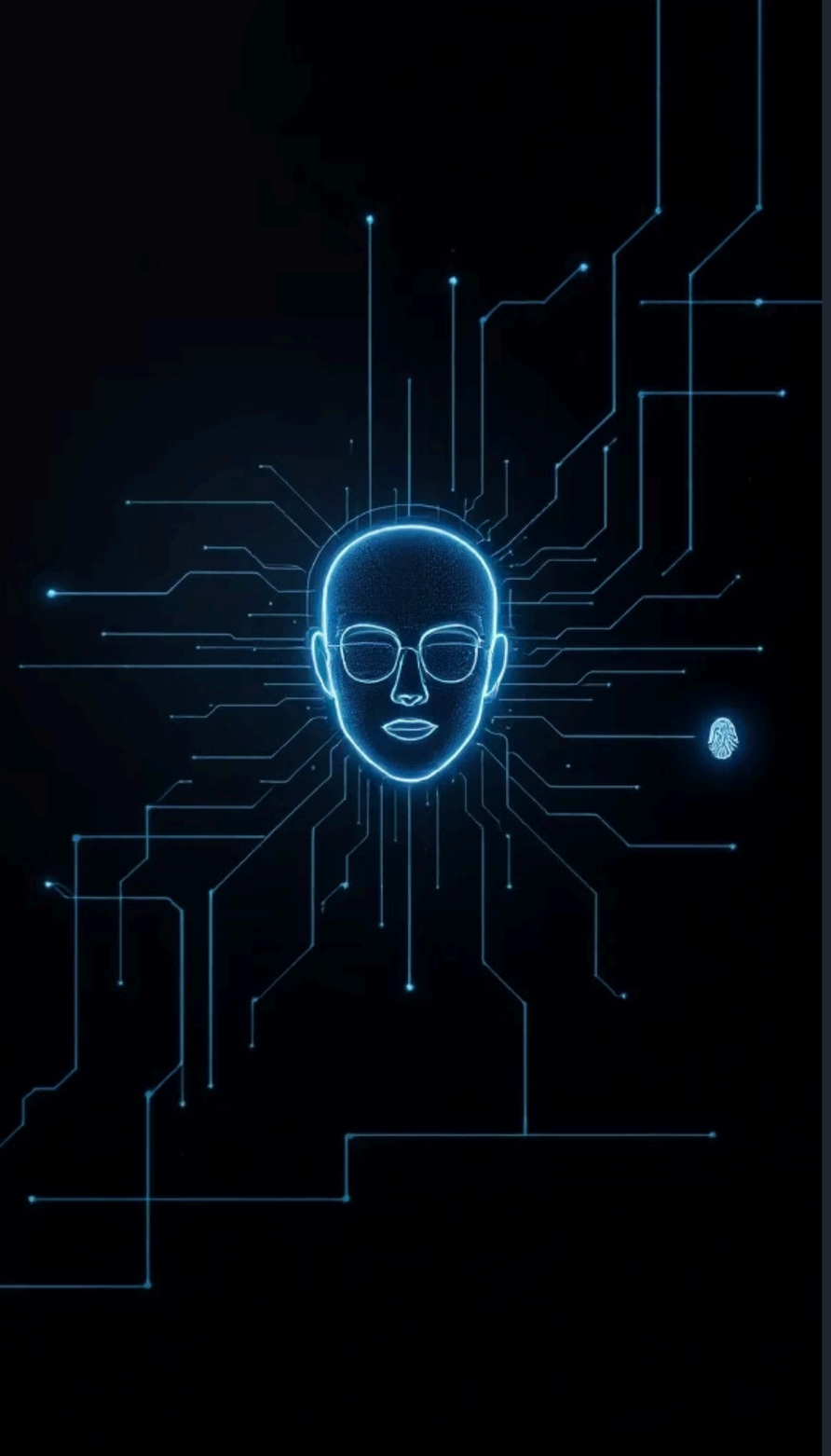
One-Time Passwords (OTPs) are better, but they can still be stolen. This happens through tricking people or using harmful software.

Fake Faces

Basic face scan systems can be tricked by photos or videos. This shows we need smarter ways to know if a face is real.

With so many clever online attacks today, we need a stronger security plan. It must have many layers to check who you are.

Made with GAMMA



Our Strong Answer: Using More Than One Way to Check Identity

Our system fixes problems with current ways to log in. It uses several methods to check who you are, along with smart security rules.

1 Checking Identity with Face and Fingerprint

We use both face scanning with special code and fingerprint checks to make things safer.

2 Smart Security Tools

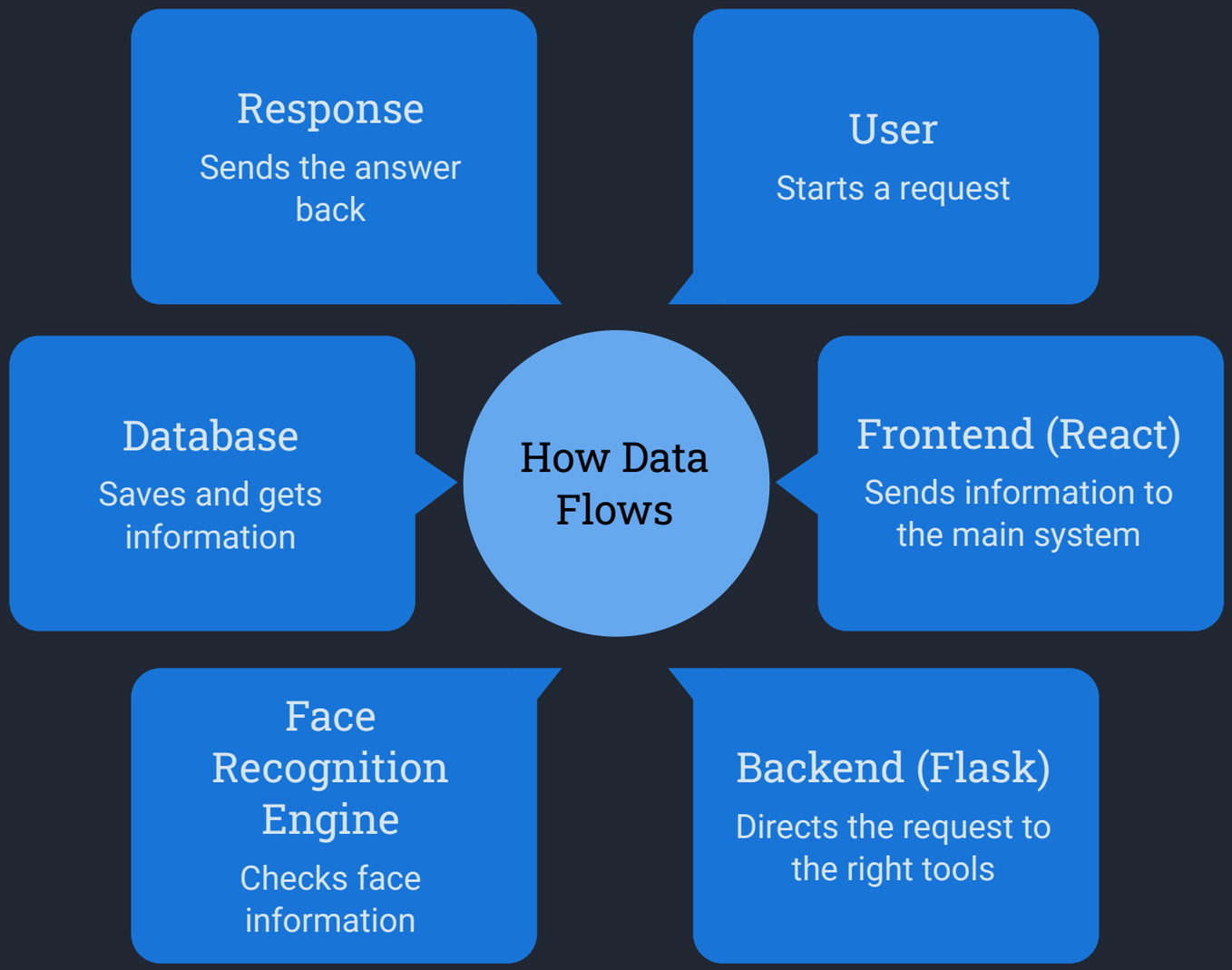
Our system includes an account lock. If someone tries to log in wrong too many times, it stops them from guessing passwords.

3 Safe Ways to Stay Logged In

We use JWT (JSON Web Tokens) to keep your online session safe. This means your login stays secure and cannot be easily stolen.

Made with GAMMA

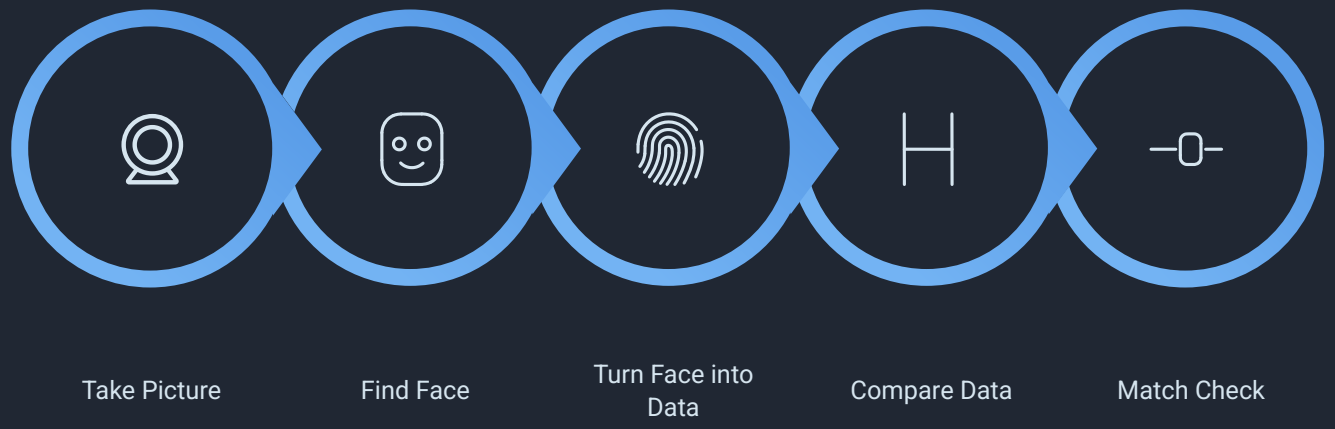
How Our System Works



This diagram shows how data moves easily from you, through our system, to special tools that check your face, and then to a safe place where it's stored.

Made with GAMMA

Face Recognition: How It Works



First, a picture is taken. Then, the system checks if it matches a saved face to confirm who you are.

Key Steps:

- **Take Picture:** A clear photo is taken from a camera.
- **Find Face:** The system finds the face in the picture.
- **Turn Face into Data:** Unique parts of the face are turned into a special code.
- **Compare Data:** This code is compared to saved face codes.
- **Match Check:** The system decides if the face matches closely enough.

This step-by-step process helps make sure it's accurate and hard to trick.

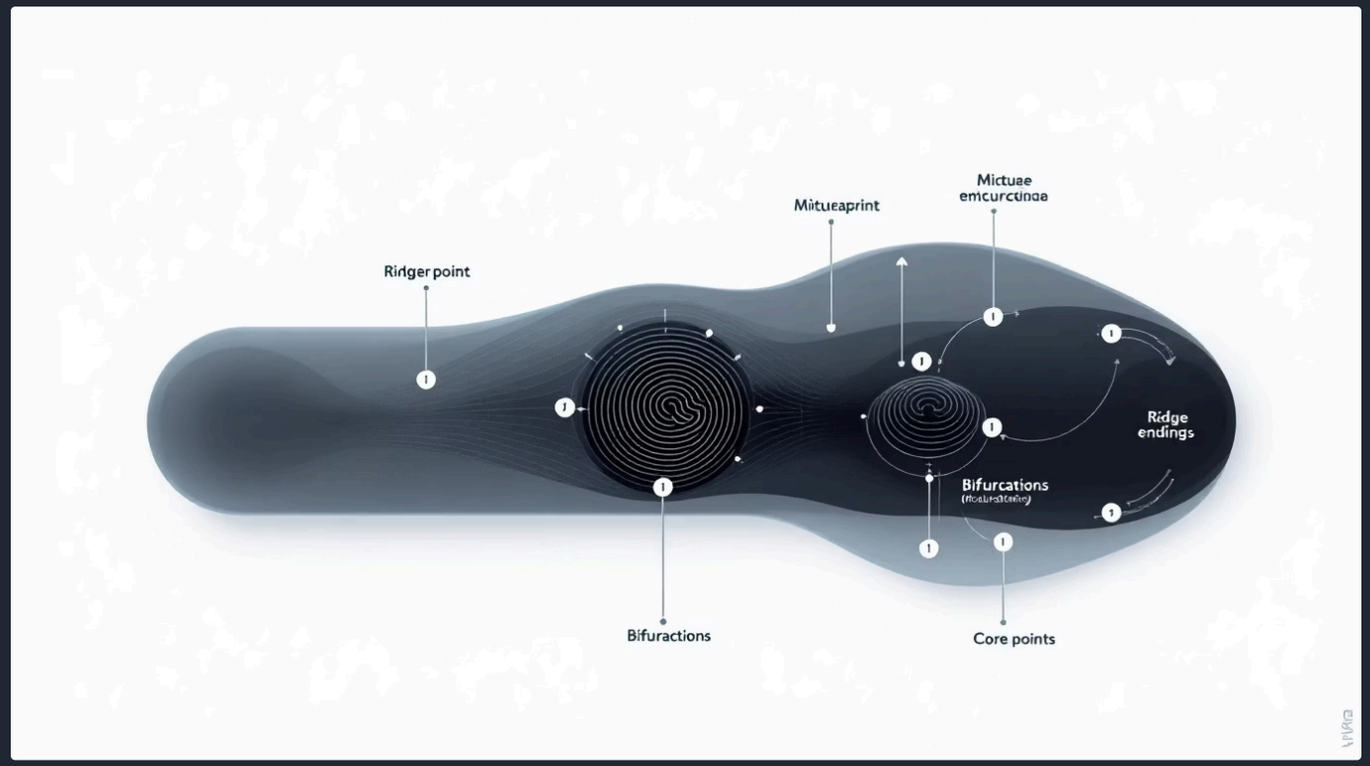
Made with GAMMA

Fingerprints: Why They're Special

How Your Fingerprint Works

Fingerprints are very good for checking who you are because each one is different and stays the same. No two people, not even twins, have the exact same fingerprints.

- **Special Lines:** These are the wavy lines and loops that make unique patterns on your fingertips.
- **Key Points:** These are tiny spots where the lines end, split, or join. They are the main things used to match fingerprints.
- **Matching System:** A safe computer program compares these tiny key points from your scanned finger to a saved record. It doesn't use the full picture of your fingerprint.



Made with GAMMA

Privacy First: We Don't Save Your Pictures

We care a lot about your privacy and keeping your data safe. Our system uses a clever way to store information.

Better Privacy for You

We only store numbers, not your actual pictures. This greatly lowers the chance of your private information getting out.

Safe Storage of Data

Your unique body data is saved as special numbers that cannot be turned back into your original image. This means others can't use them.

Can't Recreate Your Image

It's almost impossible to build your face back from these numbers. This adds a strong protection against unwanted recreation.

More Security

This method reduces dangers from data hacks. If these numbers are stolen, they can't be used to make new fake identities.

Made with GAMMA